

KOHLER Unified Enterprise AI Agent

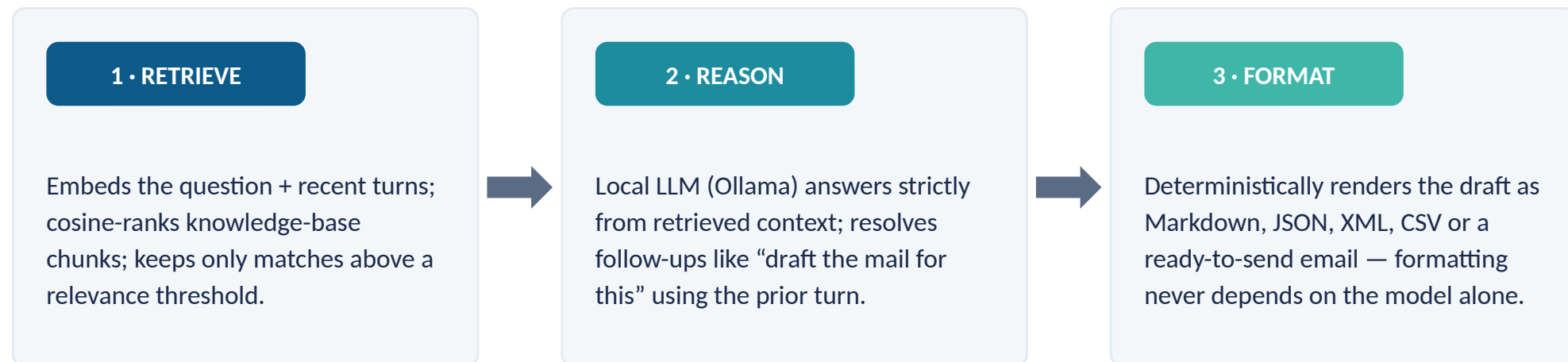
A local-first, grounded conversational agent that answers HR, finance, support and privacy questions — then formats the answer any way the business needs it.

TRACK 3 • ENTERPRISE AI AGENT

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Approach & Architecture

A three-node LangGraph pipeline keeps every answer grounded, and lets output format be decided at run time.



1 Local-first & private

Runs entirely on Ollama on one machine — no HR/finance document ever leaves the device, no API key.

2 Grounded, not guessed

Retrieval flags “NO_RELIABLE_SOURCE_MATCH” when nothing relevant exists; the model is told to say so, not invent policy.

3 Adaptive output

One conversation, many outputs: JSON for a system, an email draft for a manager, an Excel/PDF file to download.

Tech Stack & Core Capabilities

A modular Python package — nine single-purpose modules behind one Streamlit UI.

STACK

- **Orchestration:** LangGraph + LangChain-core (retrieve → reason → format state graph)
- **Local models:** Ollama runtime — qwen2.5:3b (chat) + nomic-embed-text (embeddings)
- **Retrieval:** In-memory cosine-similarity semantic index (dependency-light, swappable)
- **Interface:** Streamlit chat UI with live source citations and per-answer downloads
- **Export engine:** python-docx, openpyxl, reportlab → DOCX / XLSX / PDF / JSON / XML / CSV / ZIP

WHAT IT DOES

1

Multi-domain knowledge

HR, finance, customer support and privacy documents indexed side by side.

2

Multi-turn memory

Conversation persists locally; the agent resolves “this / that / the previous answer.”

3

Format on demand

Say “as JSON” or “draft the email” mid-chat — no need to touch a setting.

4

Self-testing

11 pytest cases + a dependency-free self-test cover every non-LLM code path.

Innovation Pitch & Impact

Why a deterministic formatting layer on top of a grounded local RAG pipeline is the differentiator.

MAPPED TO EVALUATION CRITERIA

Approach & Innovation



Grounded RAG + deterministic re-formatting layer, not prompt-and-hope output

Technical Execution



9-module package, 11 automated tests, self-test runs without any LLM

UX & Feasibility



One-command Streamlit chat; runs on a laptop behind any corporate firewall

Business & Sustainability



Zero cloud inference cost; keeps sensitive HR/finance data on-device

AT A GLANCE

4

knowledge domains: HR, finance, support, privacy

7+

output formats: MD / JSON / XML / CSV / email / DOCX / XLSX / PDF

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cloud API calls — fully local inference via Ollama